Paul He

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Education

University of Toronto — September 2021 - May 2025
Honours Bachelor of Science in Computer Science — *Toronto, Canada*

- Enrolled in Arts & Science Co-Op program
- Dean's List
- **Relevant Coursework:** Introduction to Machine Learning

Swiss Federal Institute of Technology (ETH Zürich) — September 2023 - August 2024
Exchange Year — *Zürich, Switzerland*

- EESTEC Generative AI hackathon winner
- **Relevant Coursework:** Visual Computing, Natural Language Processing (Graduate), Probabilistic Artificial Intelligence (Graduate), Machine Perception (Graduate), Large Language Models (Graduate), Advanced Formal Language Theory (Graduate)

Experience

Student Researcher — June 2024 – Present
Rycolab, Institute of Machine Learning, ETH Zürich — *Zürich, Switzerland*

- Engaged in research under the supervision of Tianyu Liu, Prof. Dr. Ryan Cotterell.
- Developed a **novel** answer attribution method for language models with concepts from **information theory**.
- Conducted experiments by implementing the proposed mechanism with **PyTorch** and training on **GPU clusters**.
- Analyzed experimental results and created synthetic data to generalize problem scenarios.
- Reviewed literature to contextualize findings and enhance the developed method.
- Prepared documentation to communicate research progress.

Projects

Word Segmentation and Part-of-Speech Tagging with Transformer — October 2023 – January 2024

- Designed a **new decoder embedding mechanism** allowing the decoder to gain access to finer-grained features, enabling it to better capture subtle linguistic nuances and dependencies between characters and tags.
- Implemented a **transformer** model using **PyTorch**, beam-search for generation.
- Achieved leading-edge results, demonstrating the model's efficiency with a validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging after just 10 epochs.

Mental Health Support Bot | *Python, beautifulsoup4, FastAPI* — October 2023

- Led the development of simple mental health chatbot web-app tailored for ETH Computer Science students, assisting team members in debugging.
- Performed **web scraping** to obtain data to perform **few-shot prompting**.
- Implemented a real-time chat feature FastAPI to host our chat platform and model on a server, allowing multiple chat's with the bot at once, and also recalls previous conversations.

Clinic Management System | *Java, JUnit, Git* — June 2022 - August 2022

- Created a Clinic Management System adhering to **Clean Architecture** and **SOLID principles**, while incorporating **Design Patterns** for optimal code structure.
- Collaborated with a **team of 8**, leveraging **Git & Github** for efficient version control, pull requests, issue tracking, and code reviews.
- Ensured code reliability through comprehensive test coverage by developing robust test suites using the JUnit framework.

Technical Skills

Programming Languages: Python, C, Java, R, SQL
Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask
Tools/Platforms: Git, Docker, Jupyter, GitHub
Databases: PostgreSQL, psycopg2